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Directions for Use

# Aminoplasmal® Paediatric B. Braun 10% Solution for infusion

## 1. NAME OF THE MEDICINAL PRODUCT

Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion

## 2. QUALITATIVE AND QUANTITATIVE COMPOSITION

The solution for infusion contains:

| | per 1 ml | per 250 ml |
|---|---|---|
| Isoleucine | 5.10 mg | 1.28 g |
| Leucine | 7.60 mg | 1.90 g |
| Lysine monohydrate (equivalent to lysine) | 9.88 mg (8.80 mg) | 2.47 g (2.20 g) |
| Methionine | 2.00 mg | 0.50 g |
| Phenylalanine | 3.10 mg | 0.78 g |
| Threonine | 5.10 mg | 1.28 g |
| Tryptophan | 4.00 mg | 1.00 g |
| Valine | 6.10 mg | 1.53 g |
| Arginine | 9.10 mg | 2.28 g |
| Histidine | 4.60 mg | 1.15 g |
| Alanine | 15.90 mg | 3.98 g |
| Glycine | 2.00 mg | 0.50 g |
| Aspartic acid | 6.60 mg | 1.65 g |
| Glutamic acid | 9.30 mg | 2.33 g |
| Proline | 6.10 mg | 1.53 g |
| Serine | 2.00 mg | 0.50 g |
| N-Acetyltyrosine (equivalent to tyrosine) | 1.30 mg (1.06 mg) | 0.33 g (0.27 g) |
| Acetylcysteine (equivalent to cysteine) | 0.700 mg (0.520 mg) | 0.175 g (0.13 g) |
| Taurine | 0.300 mg | 0.075 g |
| Total amino acids | 0.1 g | 25 g |
| Total nitrogen | 0.0152 g | 3.8 g |

For the full list of excipients, see section 6.1.

## 3. PHARMACEUTICAL FORM

Solution for infusion

Product Description:
Clear, colourless up to light yellow solution, practically free from visible particles.

| | |
|---|---|
| Energy [kJ/l (kcal/l)] | 1700 (406) |
| Theoretical osmolarity [mOsm/l] | 790 |
| Acidity (titration to pH 7.4) [mmol NaOH/l] | 23 |
| pH | approx. 6.1 |

## 4. CLINICAL PARTICULARS

### 4.1 Therapeutic indications

Supply of amino acids for parenteral nutrition in combination with energy (glucose and lipids) and electrolytes containing solutions, when oral or enteral nutrition is impossible, insufficient or contraindicated.
The solution is indicated for newborn infants, term and preterm infants and toddlers and children.

### 4.2 Posology and method of administration

Posology

*Paediatric population*

The dosages for the age group stated below are average values for guidance. The exact dosage should be adjusted individually according to age, developmental stage and prevailing disease.
Dosing should be commenced below target infusion rate value and be increased to target value within the first hour.

Parenteral amino acid supply considered adequate for most paediatric patients:

**Daily dose for preterm newborn infants:**
1.5 - 4.0 g amino acids/kg body weight ≙ 15 - 40 ml /kg body weight

**Daily dose for term newborn infants (0 – 27 days):**
1.5 - 3.0 g /kg body weight ≙ 15 - 30 ml/kg body weight.

**Daily dose for infants and toddlers (28 days to 23 months):**
1.0 - 2.5 g /kg body weight ≙ 10 - 25 ml/kg body weight

**Daily dose for children (2 to 11 years):**
1.0 - 2.0 g/kg body weight ≙ 10 - 20ml/kg body weight

Critically ill children: For critically ill patients the advisable amino acid intake may be higher (up to 3.0 g amino acids/kg body weight per day).

*Patients with renal/hepatic impairment*

The doses should be adjusted individually in patients with hepatic or renal insufficiency (see also section 4.4). Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion is contraindicated in severe hepatic insufficiency and severe renal insufficiency in absence of renal replacement therapy (see section 4.3).

*Duration of use:*

This solution can be administered as long as parenteral nutrition is indicated.

Method of administration

Intravenous use.
For central venous infusion only.
When used in infants aged from preterm to 2 years old, the solution (in bags and administration sets) should be protected from light exposure until administration is completed (see sections 4.4, 6.3 and 6.6). During preparation of mixtures a light protecting overwrap might not be suitable. Nevertheless, attention should be paid to reduce light exposure as much as possible during preparation of mixtures.

### 4.3 Contraindications

- Hypersensitivity to any of the active substance(s) or to any of the excipients listed in section 6.1
- Inborn errors of amino acid metabolism
- Severe circulation disorders with vital risk (e.g. shock)
- Hypoxia
- Metabolic acidosis
- Severe hepatic insufficiency
- Severe renal insufficiency in absence of renal replacement therapy
- Decompensated cardiac insufficiency
- Acute pulmonary oedema
- Disturbances of the electrolyte and fluid balance

### 4.4 Special warnings and precautions for use

The medicinal product should only be administered after careful benefit-risk assessment in the presence of disorders of amino acid metabolism of other origin than stated under section 4.3.

Light exposure of mixtures for intravenous parenteral nutrition, especially after admixture with trace elements and/or vitamins, may have adverse effects on clinical outcome in neonates, due to generation of peroxides and other degradation products. When used in infants aged from preterm to 2 years old, Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion should be protected from ambient light exposure until administration is completed (see sections 4.2, 6.3 and 6.6). During preparation of mixtures a light protecting overwrap might not be suitable. Nevertheless, attention should be paid to reduce light exposure as much as possible during preparation of mixtures.

Care should be exercised in the administration of large volume infusion fluids to patients with cardiac insufficiency.

Caution should be exercised in patients with increased serum osmolarity.

Disturbances of fluid and electrolyte balance (e.g. hypotonic dehydration, hyponatraemia, hypokalaemia) should be corrected prior to the administration of parenteral nutrition.

Serum electrolytes, blood glucose, fluid balance, acid-base balance and renal function should be monitored regularly.

Monitoring should also include serum protein and liver function tests.

In patients with renal insufficiency, the dose must be carefully adjusted according to individual needs, severity of organ insufficiency and the kind of instituted renal replacement therapy (haemodialysis, haemofiltration etc.).

In patients with hepatic insufficiency, the dose must be carefully adjusted according to individual needs and severity of organ insufficiency.

Amino acid solutions are only one component of parenteral nutrition. For complete parenteral nutrition, substrates for non-protein energy supply, essential fatty acids, electrolytes, vitamins, fluids and trace elements must be administered together with amino acids.

Paediatric formulations should be used in the case of micronutrients.

During long-term use (several weeks), the blood count and coagulation factors should be monitored more carefully.

### 4.5 Interaction with other medicinal products and other forms of interaction

None known.

### 4.6 Fertility, pregnancy and lactation

Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion is intended for use in the paediatric population only.

### 4.7 Effects on ability to drive and use machines

Not relevant.

### 4.8 Undesirable effects

Undesirable effects that, however, are not specifically related to the product but to parenteral nutrition in general may occur, especially at the beginning of parenteral nutrition.

Undesirable effects are listed according to their frequencies as follows:

| | |
|---|---|
| Very common | (≥ 1/10) |
| Common | (≥ 1/100 to < 1/10) |
| Uncommon | (≥ 1/1,000 to < 1/100) |
| Rare | (≥ 1/10,000 to < 1/1,000) |
| Very rare | (< 1/10,000) |
| Not known | (cannot be estimated from the available data) |

**Immune system disorders**
Not known: Anaphylactic / anaphylactoid reaction

**Gastrointestinal disorders**
Uncommon: Nausea, vomiting

**Skin and Subcutaneous Tissue Disorders**
Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

### 4.9 Overdose

*Symptoms of fluid overdose*

Overdose or too high infusion rates may lead to hyperhydration, electrolyte imbalance and pulmonary oedema.

*Symptoms of amino acid overdose*

Overdose or too high infusion rates may lead to intolerance reactions manifesting in the form of sickness, vomiting, shivering, headache, metabolic acidosis, hyperammonaemia and renal amino acid losses.

*Treatment*

If intolerance reactions occur, the amino acid infusion should be interrupted temporarily and later on resumed at a lower infusion rate.

## 5. PHARMACOLOGICAL PROPERTIES

### 5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Blood substitutes and perfusion solutions, i.v. solutions for parenteral nutrition, amino acids
ATC code: B05BA01

Mechanism of action

The aim of parenteral nutrition is the supply of all nutrients necessary for the growth, maintenance and regeneration of body tissues etc.

Amino acids are of special importance as they partly are essential for protein synthesis. Intravenously administered amino acids are incorporated in the respective intravascular and intracellular amino acid pools. Both endogenous and exogenous amino acids serve as substrate for the synthesis of functional and structural proteins.

To prevent the metabolisation of amino acids for energy production, and also to fuel the other energy-consuming processes in the organism, simultaneous non-protein energy supply (in the form of carbohydrates or fats) is necessary.

### 5.2 Pharmacokinetic properties

Absorption

Because this medicinal product is infused intravenously, the bio-availability of the amino acids contained in the solution is 100%.

Distribution

Amino acids are incorporated in a variety of proteins in different tissues of the body. In addition each amino acid is present as free amino acid in the blood and inside cells.

The composition of the amino acid solution is based upon the results of clinical investigations of the metabolism of intravenously administered amino acids. The quantities of the amino acids contained in the solution have been chosen so that a homogenous increase of the concentrations of all plasma amino acids is achieved. The physiological ratios of plasma amino acids, i.e. the amino acid homeostasis, are thus maintained during infusion of the medicinal product.

Normal foetal growth and development depend on a continuous supply of amino acids from the mother to the foetus. The placenta is responsible for the transfer of amino acids between the two circulations.

Biotransformation

Amino acids that do not enter protein synthesis are metabolised as follows: The amino group is separated from the carbon skeleton by transamination. The carbon chain is either oxidised directly to $CO_2$ or utilised as substrate for gluconeogenesis in the liver. The amino group is also metabolized in the liver to urea.

Elimination

Only minor amounts of amino acids are excreted unchanged in the urine.

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## 6. PHARMACEUTICAL PARTICULARS

### 6.1 List of excipients
Citric acid monohydrate (for pH-adjustment)
Water for injection.

### 6.2 Incompatibilities
Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion can only be mixed with other nutrients such as carbohydrates, lipids, vitamins and trace elements for which compatibility has been documented.
Compatibility data for different additives (e.g. electrolytes, trace elements, vitamins) and the corresponding shelf life of such admixtures can be provided on demand by the manufacturer. See also section 6.6.

### 6.3 Shelf life
*Unopened (bag in overwrap)*

250ml: 24 months

*After first opening*

*The medicinal product should be used immediately.*

*When used in infants aged from preterm to 2 years old, the solution (in bags and administration sets) should be protected from light exposure until administration is completed (see sections 4.2, 4.4 and 6.6).*

*During preparation of mixtures*
A light protecting overwrap during preparation of mixtures might not be suitable. Nevertheless, attention should be paid to reduce light exposure as much as possible during preparation of mixtures.

*After admixture of additives*
From a microbiological point of view, mixtures should be administered immediately after preparation. If not used immediately, storage times and conditions of mixtures prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C – 8°C, unless mixing has taken place under controlled and validated aseptic conditions.

### 6.4 Special precautions for storage
Unopened (bag in overwrap):Do not store above 30°C.
Do not freeze.

### 6.5 Nature and contents of container
Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion is supplied in flexible bags made from a multilayer foil (Polypropylene, Styrene Ethylene Butylene Styrene (SEBS) and Copolyester ether). The inner layer in contact with the solution consists of polypropylene. The bags contain 250 ml.

The bag is packed in a protective overwrap. An oxygen absorber and an oxygen indicator are placed between the bag and the overwrap; the oxygen indicator is a thermoformed blister and contains the oxygen sensitive dye resorufin sodium; the oxygen absorber sachet is made of inert material and contains iron hydroxide (Figure A).
Pack size: 12 x 250 ml.

Not all pack sizes may be marketed.

### 6.6 Special precautions for disposal and other handling
No special requirements for disposal.
Containers are for single use only. Discard overwrap, oxygen indicator, oxygen absorber, container and any unused contents after use.

Before opening the overwrap, check the colour of the oxygen indicator (see Figure A). Do not use if the oxygen indicator turned pink. Use only if the oxygen indicator is yellow.

Only to be used if the solution is clear, colourless up to light yellow, particle free and the bag and its closure are undamaged.

When used in infants aged from preterm to 2 years, parenteral nutrition solutions containing Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion should be protected from light exposure, until administration is completed. Exposure of such solutions to ambient light, especially after admixture with trace elements and/ or vitamins, generates peroxides and other degradation products that can be reduced by protection from light exposure (see sections 4.2, 4.4 and 6.3).

During preparation of mixtures a light protecting overwrap might not be suitable. Nevertheless, attention should be paid to reduce light exposure as much as possible during preparation of mixtures.

Use a sterile giving set for administration.

If in the setting of complete parenteral nutrition it is necessary to add other nutrients such as carbohydrates, lipids, vitamins, electrolytes and trace elements to this medicinal product, admixing must be performed under strict aseptic conditions. Mix well after admixture of any additive. Pay special attention to compatibility.

**Aminoplasmal® Paediatric B. Braun 10% Solution for Infusion: Handling**

**Figure A: Bag and Overwrap**

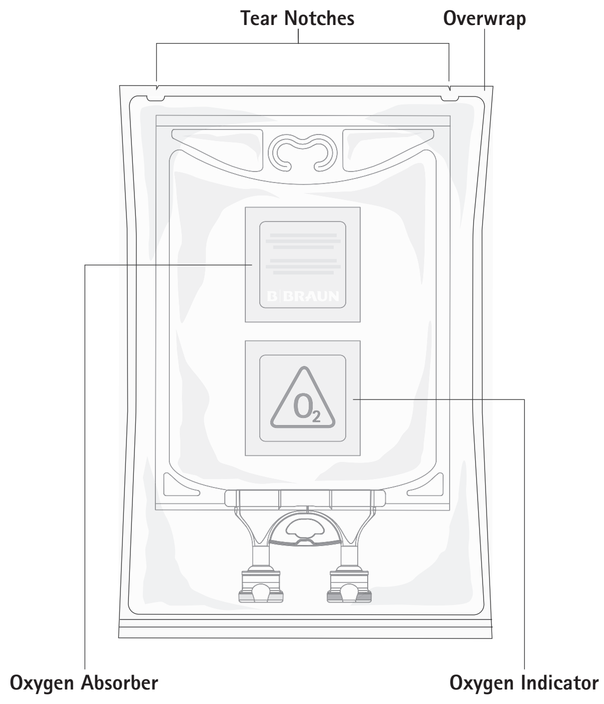
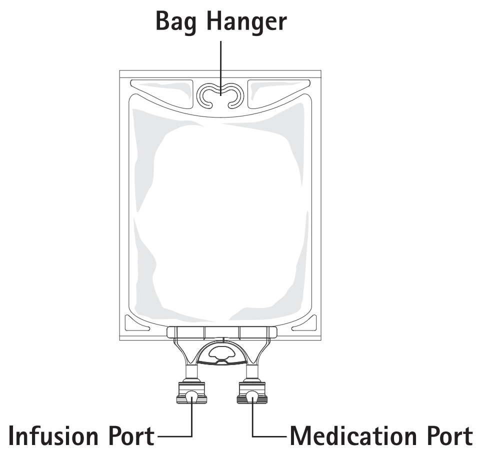
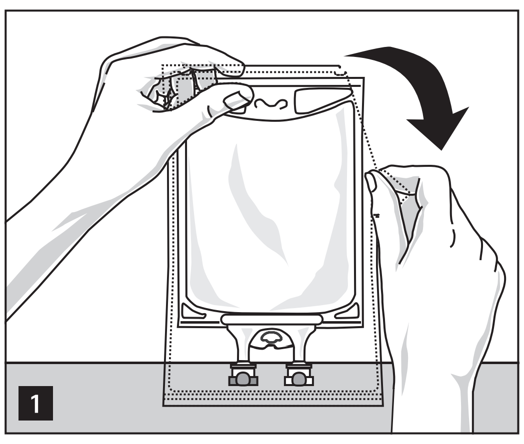


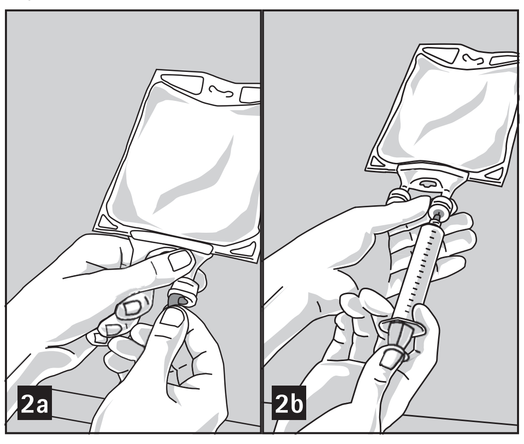

**Figure B: Bag**

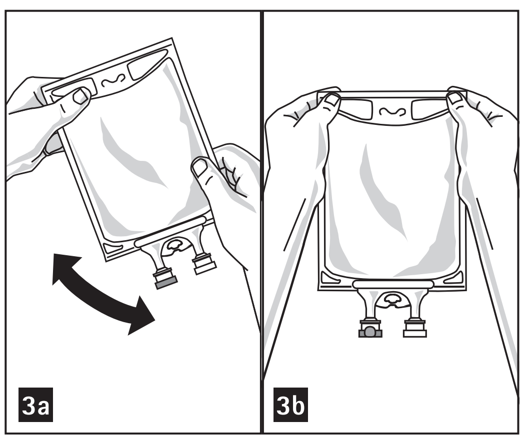



**To Open:**



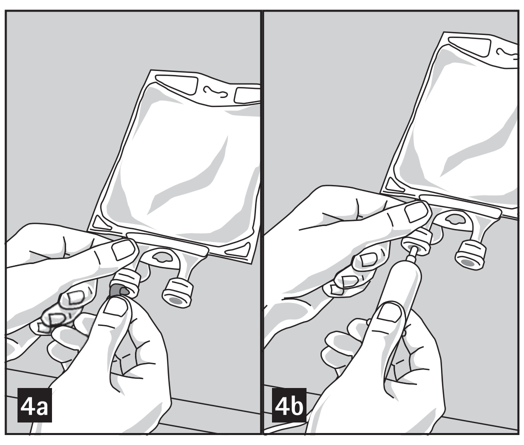

Remove the bag from its protective overwrap starting from the tear notches on top and remove solution container (figure 1).
Discard overwrap, oxygen indicator and oxygen absorber.
Check for leakages. If bag leaks, discard product as sterility may be impaired.

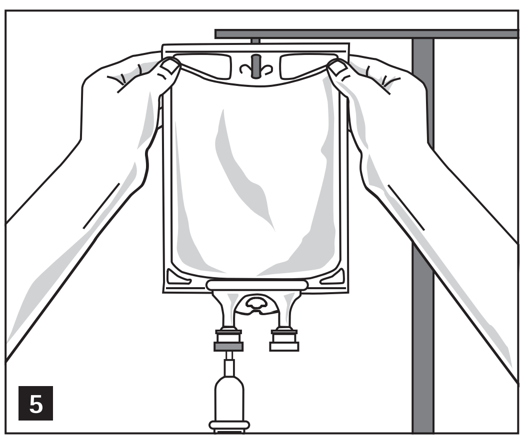



### To Add Medication:
Admixtures must be prepared following strict aseptic techniques.
Compatible supplemental medication may be added through the medication port (transparent colour).

1. Prepare medication port (transparent colour) by removal of aluminium foil (figure 2a). Please note: The area underneath the foil of the medication port is sterile.
2. Puncture resealable medication port and inject additive(s) (figure 2b).
3. Mix solution and medication thoroughly (figure 3a).
4. Medication port may be swabbed with disinfection agent (e.g. iso-propanol) before re-puncturing.
5. Check admixture visually for particulate matter (figure 3b).




**During preparation of mixtures:**

**A light protecting overwrap during preparation of mixtures might not be suitable. Nevertheless, attention should be paid to reduce light exposure as much as possible during preparation of mixtures.**

**Preparation for Administration:**

1. Remove aluminium foil of infusion port (green colour) at the bottom of container (figure 4a) and attach administration set (figure 4b): use non-vented infusion set or close air vent on a vented set. Follow directions for use of the infusion set. Please note: The area underneath the foil of the infusion port is sterile.
2. Hang bag on an i.v. pole (figure 5).

### Additional Information:
Container is PVC-free, DEHP-free, latex-free

## 7. DATE OF REVISION OF THE TEXT

**February 2024**

Product Registration Holder:
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